One complex metric and geometric adelic nefness

Abstract

We prove that an arithmetic adelic line bundle on a smooth quasi-projective variety over a number field has nef geometric part if its prescribed continuous metric at one complex embedding has semipositive first Chern current. The proof constructs semiample approximations to every positive rational ample perturbation by boundary extension, multiplier ideals, and Nadel vanishing, followed by specialization and Galois descent. This gives geometric nefness of Gao–Zhang's canonical Gross–Schoen adelic line bundle.

Keywords. Adelic line bundles; semipositive metrics; multiplier ideals; geometric nefness; Gross–Schoen cycles.

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1 Introduction

A line bundle on a smooth projective complex variety is nef if it admits a continuous Hermitian metric with semipositive curvature. For a quasi-projective variety, the corresponding question depends on the extension of the line bundle at the boundary.

In this paper, a *compactified $\mathbb{Q}$-line bundle* on a smooth quasi-projective variety U is an equivalence class of boundary-Cauchy sequences of $\mathbb{Q}$-line bundles P_i on normal projective models $U \subset X_i$, with specified identifications $\alpha_i : L_U \xrightarrow{\sim} P_i|_U$ for a fixed $\mathbb{Q}$-line bundle L_U. On a common model, the differences $P_i - P_j$ are boundary divisors, measured by upper and lower bounds in multiples of a fixed effective boundary divisor. Two sequences are equivalent if their difference tends to zero in this boundary norm, after identifying their underlying line bundles on U. These are Yuan–Zhang's geometric adelic line bundles [10]; see Definition 2.3 for details.

We prove that a continuous semipositive metric at one complex place implies geometric nefness when the metric and the compactified line bundle are limits of the same model data. This compatibility holds for arithmetic adelic line bundles. Yuan [9, Section 3.3.5] and Gao–Zhang [5, Section 4.1] each give a definition of a Gross–Schoen adelic line bundle. Applied to Gao–Zhang's line bundle, our result answers the geometric nefness question in [5, Remark 5.2].

1.1 Complex semipositivity and geometric nefness

We follow Yuan–Zhang's definitions of projective models and the boundary topology [10, Sections 2.4–2.5.1], and of strongly nef and nef adelic line bundles [10, Definition 2.5.2]; see also Section 2.

Let U be a smooth complex quasi-projective variety. Fix a smooth projective compactification $U \subset X_0$ and an effective integral Cartier divisor D_0 with $\mathrm{Supp}(D_0) = X_0 \setminus U$. We write $\|\cdot\|_{D_0}$ for the corresponding boundary norm. Comparisons of model line bundles are understood with respect to their specified identifications over U.

For a continuous Hermitian metric h on a line bundle L_U and a local frame e, write $\varphi_{h,e} = -\log\|e\|_h$. We say that h has *semipositive curvature* if $c_1(L_U, h) = dd^c\varphi_{h,e} \geq 0$ in the sense of currents; our normalization is recalled in Definition 2.5. When $D_0 \neq 0$, choose a smooth metric on $\mathcal{O}(D_0)$, normalized so that its boundary Green function $g_0 = -\log\|s_{D_0}\| \geq 1$ on U. If U is projective, take $X_0 = U$, $D_0 = 0$, and $\mathcal{L} = L_U$; in this case, the boundary conditions below are omitted.

Compactified metrics. Let $\mathcal{L}$ be a compactified $\mathbb{Q}$-line bundle on U with underlying line bundle L_U. A *compactified metric* on $\mathcal{L}$ is a continuous Hermitian metric h on L_U with the following property. For every rational $t > 0$, there exist a projective modification $p_t : X_t \to X_0$, with X_t normal and p_t an isomorphism over U, a $\mathbb{Q}$-line bundle P_t on X_t, a specified isomorphism $\alpha_t : L_U \xrightarrow{\sim} P_t|_U$, and a continuous Hermitian metric h_t on P_t, such that

$$\|P_t - \mathcal{L}\|_{D_0} \leq t, \qquad \left|\varphi_{\alpha_t^* h_t, e} - \varphi_{h,e}\right| \leq t g_0 \quad \text{on } U. \tag{1.1}$$

Both comparisons use the same specified isomorphism α_t. For projective U, every continuous Hermitian metric on L_U is a compactified metric. See Definition 2.7.

Theorem 1.1 (Complex semipositivity and geometric nefness)**.** *Let U be a smooth irreducible complex quasi-projective variety of positive dimension, and let $\mathcal{L}$ be a compactified $\mathbb{Q}$-line bundle on U with underlying line bundle L_U. If $\mathcal{L}$ admits a compactified metric h satisfying*

$$c_1(L_U, h) \geq 0,$$

then $\mathcal{L}$ is nef.

The prescribed complex metric of a Yuan–Zhang arithmetic adelic line bundle is a compactified metric on its geometric part. This follows from the arithmetic Cauchy condition [10, Section 2.5.1]; see Proposition 4.1. A continuous semipositive metric chosen independently of $\mathcal{L}$ need not satisfy this compatibility or imply nefness of the prescribed compactified extension.

We sketch the proof in Section 1.4 and give the full argument in Section 3.

1.2 The theorem over a number field

Write $\widehat{\mathrm{Pic}}(U/K)_{\mathbb{Q}}$ and $\widehat{\mathrm{Pic}}(U/\mathbb{Z})_{\mathbb{Q}}$ for the groups of compactified and arithmetic adelic $\mathbb{Q}$-line bundles, respectively. For an arithmetic adelic line bundle $\overline{\mathcal{L}}$, denote its geometric part by $\mathcal{L}^{\mathrm{geom}}$. It is obtained by taking the generic fibers of a model presentation, with their open isomorphisms; it is a compactified line bundle over K.

Theorem 1.2 (Geometric nefness over a number field)**.** *Let K be a number field, let U/K be a smooth geometrically integral quasi-projective variety of positive dimension, and let $\mathcal{L} \in \widehat{\mathrm{Pic}}(U/K)_{\mathbb{Q}}$ have underlying line bundle L_U. If, for an embedding $\sigma : K \hookrightarrow \mathbb{C}$, $\mathcal{L}_\sigma$ admits a compactified metric h satisfying*

$$c_1(L_{U,\sigma}, h) \geq 0,$$

then $\mathcal{L}$ is nef.

In particular, the geometric part of an arithmetic adelic line bundle is nef if its prescribed continuous metric at one complex embedding has semipositive first Chern current.

The proof in Section 4 specializes generating sections and boundary comparison maps to a finite extension of K, then descends their tensor products over the Galois conjugates. Division by the extension degree preserves the boundary error.

1.3 Gross–Schoen line bundles

For a smooth projective curve C of genus $g \geq 2$ and a degree-one class ξ with $(2g-2)\xi = \omega_C$, let Δ_ξ denote the canonical Gross–Schoen modified diagonal cycle on C^3.

For a smooth family $\pi : \mathcal{C} \to S$, Yuan constructs the admissible adelic dualizing bundle $\overline{\omega}_a$ and the adelic divisor Φ_S associated with the local φ-invariant. His Gross–Schoen line bundle is

$$\overline{G}^{Y}_{\mathcal{C}/S} = \frac{2g+1}{2g-2}\langle \overline{\omega}_a, \overline{\omega}_a \rangle_\pi - \overline{\mathcal{O}}(\Phi_S).$$

Its point heights are the canonical Gross–Schoen heights. Yuan asks whether this adelic line bundle is nef and, when the family has maximal variation, whether it is big [9, Section 3.3.5]. Here nefness and bigness are arithmetic; see Definition 2.11 for the normalization.

Gao–Zhang construct a Gross–Schoen adelic line bundle on the parameter space of canonically centered curves by a Deligne pairing. Let M be a smooth geometrically integral fine-level parameter variety over a number field K for pairs (C, ξ) with $g(C) = g \geq 3$ and $(2g-2)\xi = \omega_C$. To obtain M, start with a fine moduli scheme of smooth genus-g curves

with level structure of order at least 3, take the finite étale cover parametrizing classes ξ in the relative Picard scheme satisfying $(2g-2)\xi = \omega$, and choose a geometrically integral component over a number field of definition. Let $\pi : \mathcal{C} \to M$ be the universal curve, $J = \mathrm{Jac}(\mathcal{C}/M)$, and $i_\xi : \mathcal{C} \to J$ the Abel–Jacobi map. If $\overline{\mathcal{P}}$ is their canonically metrized Poincaré line bundle, set

$$\overline{\mathcal{Q}} = (i_\xi, i_\xi)^*\overline{\mathcal{P}}, \qquad \overline{G}_g = \langle \overline{\mathcal{Q}}, \overline{\mathcal{Q}}, \overline{\mathcal{Q}} \rangle_{\mathcal{C}\times_M \mathcal{C}/M}. \tag{1.2}$$

We use the normalization of [5, Section 4.1, equation (4.2)], specified in Definition 2.12. Gao–Zhang prove that $\overline{G}_g$ is integrable, that its point height is $\langle \Delta_\xi, \Delta_\xi \rangle_{\mathrm{BB}}$, and that its prescribed complex metric has curvature

$$c_1(\overline{G}_{g,\sigma}) = \beta_{\mathrm{GS}} \geq 0$$

[5, Theorems 4.1–4.2]. Their Remark 5.2 asks for nefness of the geometric part, denoted $\widetilde{\mathscr{L}}$ there. Applying Theorem 1.2 gives the following result.

Theorem 1.3 (Geometric nefness of the Gross–Schoen line bundle)**.** *Let $g \geq 3$, let K be a number field, and let M/K be a smooth geometrically integral fine-level parameter variety for canonically centered curves as above. The geometric part G_g^{geom} of the arithmetic adelic $\mathbb{Q}$-line bundle $\overline{G}_g$ defined by (1.2) is nef.*

1.4 Sketch of the proof of Theorem 1.1

We adapt the multiplier-ideal argument of Botero–Burgos Gil–Holmes–de Jong [2, Section 5]. The boundary estimates below give convergence to the prescribed compactified line bundle while retaining its restriction over U.

Extension across the boundary. Fix an ample $\mathbb{Q}$-line bundle A on X_0 and a rational $\varepsilon > 0$, and choose a sufficiently small rational $t > 0$. Choose approximation data $(p_t : X_t \to X_0, P_t, \alpha_t, h_t)$ as in (1.1), with X_t smooth. Set $D_t = p_t^* D_0$ and $B_t = P_t + tD_t$. Using the specified identification over U and the canonical section of the boundary divisor, we transport the metric h to $B_t|_U$. Its local weights w_t satisfy

$$2t \log |f_{D_t}| - C \leq w_t \leq C,$$

where f_{D_t} is a local equation of D_t and C is a constant on each sufficiently small chart. The upper bound allows the metric to extend across the boundary as a singular semipositive metric. Moreover, the boundary approximation gives $0 \leq B_t - \mathcal{L} \leq 2tD_0$ in the completed boundary order; see Proposition 3.2.

Multiplier ideals. The lower bound for w_t gives, for every sufficiently divisible positive integer m, the inclusions

$$\mathcal{O}(-2mtD_t) \subseteq \mathcal{J}(mw_t) \subseteq \mathcal{O}.$$

By coherence and GAGA, the multiplier ideal algebraizes to a coherent ideal $\mathcal{I}_{t,m}$ on X_t. Let $\rho : Y \to X_t$ be a principalization of this ideal, with image ideal $\mathcal{I}_{t,m}\mathcal{O}_Y = \mathcal{O}_Y(-F)$. The preceding inclusions yield

$$0 \leq \frac{F}{m} \leq 2t\rho^* D_t.$$

This is an inequality of $\mathbb{Q}$-Cartier divisors, including exceptional multiplicities; see Proposition 3.5.

Global generation. Choose an ample $\mathbb{Q}$-line bundle A_t on X_t with a specified identification $A_t|_U \simeq A|_U$ and satisfying

$$-\lambda D_t \leq A_t - p_t^* A \leq \lambda D_t$$

for an arbitrarily small rational $\lambda > 0$. Equip A_t with a smooth Hermitian metric of strictly positive curvature. The induced singular metric on $B_t + \varepsilon A_t$ then has strictly positive curvature.

Nadel vanishing and Castelnuovo–Mumford regularity [2, Theorems 5.9–5.10 and Corollary 5.11] imply that

$$\mathcal{O}(m(B_t + \varepsilon A_t)) \otimes \mathcal{I}_{t,m}$$

is globally generated for all sufficiently large divisible integers m. Here X_t and A_t are fixed before m is chosen; see Lemma 3.7.

Semiample approximation. Choose a sufficiently large divisible m and principalize $\mathcal{I}_{t,m}$ as above. Pulling back the generating sections and using the principalized image ideal gives a semiample $\mathbb{Q}$-line bundle

$$N_{t,\varepsilon,m} = \rho^*(B_t + \varepsilon A_t) - \frac{F}{m}$$

on Y, with the specified restriction $N_{t,\varepsilon,m}|_U \simeq L_U + \varepsilon A|_U$.

For $\lambda \leq t/(1+\varepsilon)$, the boundary estimates give

$$\left\| N_{t,\varepsilon,m} - (\mathcal{L} + \varepsilon A) \right\|_{D_0} \leq 2t + \varepsilon\lambda \leq 3t.$$

Letting $t \to 0$ shows that $\mathcal{L} + \varepsilon A$ is strongly nef. Taking $\varepsilon = 1/a$ and multiplying by a gives strong nefness of $a\mathcal{L} + A$ for every integer $a \geq 1$. Since A is fixed and strongly nef, $\mathcal{L}$ is nef.

Remark 1.4 (Semiample approximation). The proof also gives semiample approximations with arbitrary boundary error. Under the hypotheses of Theorem 1.1, fix an ample $\mathbb{Q}$-line bundle A on X_0. For every rational $\varepsilon > 0$ and every real $\delta > 0$, there exist a projective modification $q : Y \to X_0$, with Y smooth and q an isomorphism over U, and a semiample $\mathbb{Q}$-line bundle N on Y, together with a specified isomorphism

$$L_U + \varepsilon A|_U \xrightarrow{\sim} N|_U,$$

such that

$$\left\| N - (\mathcal{L} + \varepsilon A) \right\|_{D_0} < \delta. \tag{1.3}$$

Consequently, $\mathcal{L} + \varepsilon A$ is strongly nef for every rational $\varepsilon > 0$, and $a\mathcal{L} + A$ is strongly nef for every integer $a \geq 1$, with the same fixed ample line bundle A. If U is projective, one may take $Y = X_0 = U$ and $N = L_U + \varepsilon A$, with no boundary estimate needed.

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Use of generative AI

Generative AI was used in the preparation and revision of this manuscript. An account of this assistance and selected development records are available in the supplementary repository:

`https://github.com/DongKaicheng/one-complex-metric-ai-development`.

The release of 6 November 2026 includes the English AI-assisted development report (PDF) and the selected mathematical work-process archive (ZIP).

2 Adelic line bundles and complex metrics

We use the conventions of [10] and additive notation for tensor products of line bundles.

2.1 Compactified line bundles on quasi-projective varieties

Let k be a field of characteristic zero, and let U/k be a smooth geometrically integral quasi-projective variety. Fix a smooth projective compactification $U \subset X_0$ and an effective integral Cartier divisor D_0 with $\operatorname{Supp} D_0 = X_0 \setminus U$. When U is projective, we take $X_0 = U$ and $D_0 = 0$.

Definition 2.1 (Projective models and modifications)**.** A projective model of U is a normal projective variety X equipped with a specified dense open immersion $U \subset X$. We work with projective models dominating X_0. A *projective modification* $p : X \to X_0$ is a projective birational morphism that restricts to the identity on U. For such a modification, we write

$$D_X = p^* D_0, \tag{2.1}$$

where the pullback retains all Cartier multiplicities.

For a fixed $\mathbb{Q}$-line bundle L_U, a model presentation is a triple (X, P, α_P), where P is a $\mathbb{Q}$-line bundle on X and $\alpha_P : L_U \xrightarrow{\sim} P|_U$ is a specified isomorphism.

For two model presentations (X, P, α_P) and (X', Q, α_Q) of L_U, choose a common projective model $U \subset W$, with modifications $r_P : W \to X$ and $r_Q : W \to X'$. Their difference is represented by the $\mathbb{Q}$-Cartier divisor

$$\Delta(Q, P) = \operatorname{div}(r_Q^* \alpha_Q \circ (r_P^* \alpha_P)^{-1}), \tag{2.2}$$

which is supported on $W \setminus U$.

Definition 2.2 (Boundary norm)**.** Suppose $D_0 \neq 0$. For two model presentations P and Q as above, let $U \subset W$ be a common projective model, with induced modification $\rho : W \to X_0$. Their boundary distance is

$$\|Q - P\|_{D_0} = \inf\{c \in \mathbb{Q}_{>0} : -c\rho^* D_0 \leq \Delta(Q, P) \leq c\rho^* D_0\}. \tag{2.3}$$

The inequalities are inequalities of $\mathbb{Q}$-Cartier divisors. This norm is independent of the chosen common model; see [10, Section 2.4].

Definition 2.3 (Compactified line bundles)**.** A compactified $\mathbb{Q}$-line bundle $\mathcal{L}$ on U/k with underlying line bundle L_U is represented by a boundary-Cauchy sequence of model presentations of L_U. Two sequences represent the same compactified line bundle if there is an isomorphism between their underlying line bundles over U for which the boundary norms of their differences tend to zero.

We denote the group of compactified $\mathbb{Q}$-line bundles by $\widehat{\operatorname{Pic}}(U/k)_{\mathbb{Q}}$ and identify a model presentation (X, P, α_P) with its constant class, denoting both by P. When U is projective, we set $\widehat{\operatorname{Pic}}(U/k)_{\mathbb{Q}} = \operatorname{Pic}(U)_{\mathbb{Q}}$, so that $\mathcal{L} = L_U$.

Thus $\widehat{\mathrm{Pic}}(U/k)_{\mathbb{Q}}$ is the group of adelic line bundles in the geometric setting of Yuan–Zhang [10, Sections 1.5 and 2.5.1].

When $D_0 \neq 0$, the boundary norm extends to completed differences with a specified identification over U. We use the order defined by the closure of the cone of effective boundary divisors. Thus

$$\|v\|_{D_0} \leq c \quad \Longleftrightarrow \quad -cD_0 \leq v \leq cD_0, \tag{2.4}$$

where v is a completed boundary difference and $c \in \mathbb{Q}_{\geq 0}$. Here D_0 denotes its constant class with the canonical trivialization over U.

Definition 2.4 (Nefness)**.** Following [10, Definition 2.5.2], a compactified line bundle is *strongly nef* if it is a boundary limit of nef projective model line bundles.

A compactified line bundle $\mathcal{L}$ is *nef* if there exists a fixed strongly nef compactified line bundle $\mathcal{A}$ such that $a\mathcal{L} + \mathcal{A}$ is strongly nef for every integer $a \geq 1$.

A compactified line bundle is *integrable* if it is a difference of two strongly nef compactified line bundles.

2.2 Complex metrics and multiplier ideals

In this subsection the varieties are defined over $\mathbb{C}$. We use the standard terminology for singular Hermitian metrics and plurisubharmonic functions, as in [4, 3].

Definition 2.5 (Metric and curvature conventions)**.** Let h be a possibly singular Hermitian metric with locally integrable weights on a holomorphic line bundle L. For a nonvanishing local holomorphic frame e, we define its local weight by $\varphi_{h,e} = -\log \|e\|_h$.

We use the convention $dd^c = \frac{i}{\pi}\partial\bar{\partial}$ so that the first Chern current of (L, h) is

$$c_1(L, h) = dd^c \varphi_{h,e}.$$

The metric h is called *semipositive* if its first Chern current $c_1(L, h)$ is a positive $(1, 1)$-current; we write $c_1(L, h) \geq 0$. For singular semipositive metrics, local weights are understood in their plurisubharmonic (psh) representatives.

For $\mathbb{Q}$-line bundles, these conventions are defined by passing to an integral tensor multiple and dividing by that multiple.

Definition 2.6 (Boundary Green function)**.** Suppose $D_0 \neq 0$. Choose a smooth Hermitian metric on $\mathcal{O}(D_0)$ and rescale it by a positive constant so that

$$g_0 = -\log \|s_{D_0}\| \geq 1 \qquad \text{on } U, \tag{2.5}$$

where s_{D_0} is the canonical section of $\mathcal{O}(D_0)$.

On a projective model $U \subset X$ equipped with a modification $p : X \to X_0$, let $g_X = p^* g_0$. Locally, choose a regular frame e_{D_X} of $\mathcal{O}(D_X)$ and write $s_{D_X} = f_{D_X} e_{D_X}$. Then

$$g_X = -\log |f_{D_X}| + \gamma_X, \tag{2.6}$$

where γ_X is smooth.

Definition 2.7 (Compactified metrics)**.** Let $\mathcal{L}$ be a compactified $\mathbb{Q}$-line bundle on a smooth complex quasi-projective variety U, with underlying line bundle L_U. A *compactified metric* on $\mathcal{L}$ is a continuous Hermitian metric h on L_U admitting continuous metrized model presentations satisfying (1.1), with respect to the fixed boundary data X_0, D_0, g_0. The algebraic and metric comparisons must use the same specified isomorphisms over U; no positivity is required of the approximating metrics h_t. If U is projective, every continuous Hermitian metric on L_U is a compactified metric.

The condition is independent of the boundary data. On a common modification, two effective boundary divisors with full boundary support bound one another up to positive rational constants, as do their normalized Green functions. Shrinking the tolerance therefore transfers (1.1) from one choice to the other. Pullback to a resolution that is an isomorphism over U preserves both estimates, so the approximating modifications may be taken to have smooth sources.

Definition 2.8 (Multiplier ideal sheaf)**.** Let w denote the local psh weights of a singular Hermitian metric on a line bundle over a smooth complex variety X. For a positive integer m, define

$$\mathcal{J}(mw)_x = \left\{ a \in \mathcal{O}_{X^{\mathrm{an}},x} : |a|^2 e^{-2mw} \text{ is locally integrable near } x \right\}. \tag{2.7}$$

Here integrability is understood with respect to Lebesgue measure in local holomorphic coordinates, or equivalently, any smooth positive volume form.

The local ideals glue to a coherent analytic ideal sheaf $\mathcal{J}(mw) \subseteq \mathcal{O}_{X^{\mathrm{an}}}$, called the *multiplier ideal sheaf* [3, Proposition 5.7]. The local definitions agree on overlaps because the weights differ by locally bounded functions.

2.3 Arithmetic adelic line bundles

Definition 2.9 (Arithmetic adelic line bundles)**.** Let K be a number field and let U/K be as above. We write $\widehat{\mathrm{Pic}}(U/\mathbb{Z})_{\mathbb{Q}}$ for the arithmetic adelic $\mathbb{Q}$-line bundles of [10, Section 2.5]. Such a line bundle is represented by a Cauchy sequence of Hermitian line bundles on projective arithmetic models.

More precisely, relative to a strictly effective arithmetic boundary $\overline{\mathscr{D}}_0$, the comparisons in a cofinal model presentation satisfy

$$-\epsilon_i \overline{\mathscr{D}}_0 \le \widehat{\mathrm{div}}(\alpha_j \alpha_i^{-1}) \le \epsilon_i \overline{\mathscr{D}}_0 \qquad (j \ge i), \tag{2.8}$$

for positive rational numbers $\epsilon_i \to 0$. These are arithmetic divisor inequalities, including their Green-function components.

Definition 2.10 (Geometric part and complex metric)**.** For $\overline{\mathcal{L}} \in \widehat{\mathrm{Pic}}(U/\mathbb{Z})_{\mathbb{Q}}$, its *geometric part* $\mathcal{L}^{\mathrm{geom}} \in \widehat{\mathrm{Pic}}(U/K)_{\mathbb{Q}}$ is obtained by taking the generic fibers of a model presentation, retaining the specified isomorphisms over U.

For every embedding $\sigma : K \hookrightarrow \mathbb{C}$, the same arithmetic presentation determines a continuous Hermitian metric h_σ on the underlying line bundle over U_σ. These constructions are independent of the presentation; see [10, Section 2.5.5 and Proposition 3.4.1].

Arithmetic nefness is understood in the sense of [10, Section 2.5]. In particular, it implies nefness of the geometric part. The converse is not asserted: the geometric nefness considered in this paper does not include the arithmetic degree conditions.

2.4 The Gross–Schoen line bundles

We recall the definitions of Gross–Schoen adelic line bundles given by Yuan [9, Section 3.3.5] and Gao–Zhang [5, Section 4.1].

Definition 2.11 (Yuan's Gross–Schoen line bundle)**.** Let S be a smooth geometrically integral quasi-projective variety over a number field, and let $\pi : \mathcal{C} \to S$ be a smooth

projective family of curves of genus $g \geq 2$. Write $\overline{\omega}_a$ and $\overline{\mathcal{O}}(\Delta)_a$ for the canonical admissible adelic extensions of the relative dualizing and diagonal line bundles. Set

$$\overline{\mathcal{O}}(\Phi_S) = \langle \overline{\omega}_a, \overline{\omega}_a \rangle_\pi - \langle \overline{\mathcal{O}}(\Delta)_a, \overline{\mathcal{O}}(\Delta)_a, \overline{\mathcal{O}}(\Delta)_a \rangle_{\pi^{(2)}}, \tag{2.9}$$

where $\pi^{(2)} : \mathcal{C} \times_S \mathcal{C} \to S$. Its underlying line bundle has a canonical trivialization induced by adjunction.

In this notation, Yuan's Gross–Schoen adelic line bundle is

$$\overline{G}^Y_{\mathcal{C}/S} = \frac{2g+1}{2g-2} \langle \overline{\omega}_a, \overline{\omega}_a \rangle_\pi - \overline{\mathcal{O}}(\Phi_S). \tag{2.10}$$

Here $\overline{\mathcal{O}}(\Phi_S)$ is the adelic line bundle associated with the relative φ-invariant in [9, Section 3.3.2]; the Gross–Schoen combination (2.10) is introduced in [9, Section 3.3.5].

Definition 2.12 (Gao–Zhang's Gross–Schoen line bundle)**.** Let S be a smooth quasi-projective variety over a number field, and let $\pi : \mathcal{C} \to S$ be a smooth projective family of curves of genus $g \geq 3$. Suppose that the family is equipped with a relative degree-one class $\xi \in \mathrm{Pic}^1(\mathcal{C}/S)$ satisfying

$$(2g-2)\xi = \omega_{\mathcal{C}/S}.$$

Following [5, Section 4.1], let $J = \mathrm{Jac}(\mathcal{C}/S)$ and $i_\xi : \mathcal{C} \to J$ be the Abel–Jacobi morphism. Let $\overline{\mathcal{P}}^\Delta$ be the canonical adelic extension of the diagonal Poincaré line bundle on J, characterized by

$$[2]^* \overline{\mathcal{P}}^\Delta = 4 \overline{\mathcal{P}}^\Delta.$$

Writing $m, p_1, p_2 : J \times_S J \to J$ for the addition morphism and the two projections, respectively, define

$$2\overline{\mathcal{P}} = m^* \overline{\mathcal{P}}^\Delta - p_1^* \overline{\mathcal{P}}^\Delta - p_2^* \overline{\mathcal{P}}^\Delta, \qquad \overline{\mathcal{Q}} = (i_\xi, i_\xi)^* \overline{\mathcal{P}}. \tag{2.11}$$

Gao–Zhang's Gross–Schoen adelic line bundle is

$$\overline{G}_g = \langle \overline{\mathcal{Q}}, \overline{\mathcal{Q}}, \overline{\mathcal{Q}} \rangle_{\pi^{(2)}}, \tag{2.12}$$

where $\pi^{(2)} : \mathcal{C} \times_S \mathcal{C} \to S$. We denote its geometric part by G_g^{geom} and its underlying line bundle on S by $G_{g,S}$.

This is the pullback of Gao–Zhang's construction on the moduli stack of pairs (C, ξ) [5, Section 4.1].

The height and curvature of this metric are recalled in Proposition 5.1.

3 Proof of Theorem 1.1

We prove Theorem 1.1 together with the semiample approximation statement of Remark 1.4. Throughout this section, the varieties are defined over $\mathbb{C}$ and the conventions of Section 2.1 apply. Since the compactified metric is continuous and has semipositive curvature, its local weights are continuous psh functions.

3.1 Extension across the boundary

Theorem 3.1 (Removable singularities for plurisubharmonic functions)**.** *Let V be a connected complex manifold and $Z \subset V$ a proper closed analytic subset. A psh function on $V \setminus Z$ that is locally bounded above near Z has a unique psh extension, obtained by upper limiting values. The extension is locally integrable unless identically $-\infty$.*

Moreover, adding a pluriharmonic function on V commutes with this extension. This is the removable-singularity theorem of [4, Chapter I, Theorem 5.24].

Proposition 3.2. *Assume that $D_0 \neq 0$ and that $(\mathcal{L}, h)$ satisfies the hypotheses of Theorem 1.1. For a rational $t > 0$, choose a projective modification $p_t : X_t \to X_0$ with X_t smooth, together with P_t, h_t, α_t satisfying (1.1), and put*

$$D_t = p_t^* D_0, \qquad B_t = P_t + tD_t.$$

Identify $B_t|_U$ with L_U using α_t and the canonical section of tD_t.

Then h extends to a singular semipositive metric on B_t. Moreover,

$$0 \leq B_t - \mathcal{L} \leq 2tD_0 \tag{3.1}$$

in the completed boundary order.

On every sufficiently small regular chart of X_t, the local weight w_t of the extended metric satisfies

$$2t \log |f_{D_t}| - C \leq w_t \leq C, \tag{3.2}$$

where f_{D_t} is a local equation of D_t and C is a constant depending on the chart.

Proof. Locally, choose regular frames of P_t and $\mathcal{O}(D_t)$, and let f_{D_t} be the local equation of D_t. Write φ and β_t for the weights of h and h_t, respectively, in the corresponding frames. The metric induced by h on $B_t|_U$ has weight

$$w_t = \varphi + t \log |f_{D_t}|. \tag{3.3}$$

Since $g_{X_t} = -\log |f_{D_t}| + \gamma_t$, condition (1.1) gives

$$\beta_t + 2t \log |f_{D_t}| - t\gamma_t \leq w_t \leq \beta_t + t\gamma_t.$$

As β_t and γ_t are locally bounded, (3.2) follows. In particular, w_t is locally bounded above near the boundary. By Theorem 3.1, it extends uniquely as a psh function across D_t. These local extensions glue by uniqueness, giving the required singular semipositive metric on B_t.

Finally, (1.1) implies $-tD_0 \leq P_t - \mathcal{L} \leq tD_0$ in the completed boundary order. Adding tD_0 yields (3.1). □

3.2 Ample perturbations on a projective model

For a projective morphism $p : X \to X_0$, twisting a p-ample line bundle by the pullback of a line bundle on X_0 preserves relative ampleness. If H is p-ample and A is ample on X_0, then $H + np^*A$ is ample for all sufficiently large integers n [8, Tags 01VJ and 0892].

Lemma 3.3 (Ample perturbation)**.** *Let $p : X \to X_0$ be a projective modification with X smooth, and let A be an ample $\mathbb{Q}$-line bundle on X_0. For every rational $\lambda > 0$, there exists an ample $\mathbb{Q}$-line bundle A_X on X, equipped with an identification $A_X|_U \simeq A|_U$, such that*

$$-\lambda D_X \leq A_X - p^*A \leq \lambda D_X. \tag{3.4}$$

Here the inequalities are inequalities of boundary $\mathbb{Q}$-Cartier divisors determined by the specified identification over U.

Proof. Choose an integral p-ample line bundle H on X. Since X_0 is smooth, the line bundle $H|_U$ extends to a line bundle E on X_0 [8, Tag 0BD9].

Then $R = H - p^*E$ is p-ample and has a specified trivialization over U. Thus R is represented by a Cartier divisor supported on $X \setminus U$. Since D_X has positive multiplicity along every boundary prime divisor, there exists $c > 0$ such that

$$-cD_X \leq R \leq cD_X.$$

By relative ampleness, $R + np^*A$ is ample for every sufficiently large divisible integer n. Choose such an n with $c/n \le \lambda$, and set

$$A_X = p^*A + \frac{1}{n}R.$$

Then A_X is ample, its restriction to U is identified with $A|_U$, and

$$-\lambda D_X \le \frac{1}{n}R = A_X - p^*A \le \lambda D_X.$$

□

For the projective models $U \subset X_t$ appearing in the proof of Theorem 1.1, we denote the corresponding ample perturbations by A_t.

3.3 Multiplier ideals and principalization

Fix a rational $t > 0$ and the data of Proposition 3.2. For a coherent ideal $\mathcal{I} \subseteq \mathcal{O}_X$ and a morphism $\rho : Y \to X$, write $\mathcal{I}\mathcal{O}_Y = \mathrm{im}(\rho^*\mathcal{I} \to \mathcal{O}_Y)$ for the image ideal.

Proposition 3.4 (Algebraization and principalization of the multiplier ideal)**.** *Choose a positive integer m such that mB_t is integral and $mt \in \mathbb{Z}$, and set $b_m = 2mt$. The multiplier ideal $\mathcal{J}(mw_t)$ algebraizes to a nonzero coherent ideal $\mathcal{I}_{t,m} \subseteq \mathcal{O}_{X_t}$ satisfying*

$$\mathcal{O}_{X_t}(-b_m D_t) \subseteq \mathcal{I}_{t,m} \subseteq \mathcal{O}_{X_t}, \qquad \mathcal{I}_{t,m}|_U = \mathcal{O}_U. \tag{3.5}$$

Moreover, there exists a projective modification $\rho_{t,m} : Y_{t,m} \to X_t$, with $Y_{t,m}$ smooth and $\rho_{t,m}$ an isomorphism over U, such that

$$\mathcal{I}_{t,m}\mathcal{O}_{Y_{t,m}} = \mathcal{O}_{Y_{t,m}}(-F_{t,m}), \tag{3.6}$$

where $F_{t,m}$ is an effective Cartier divisor supported on $Y_{t,m} \setminus U$.

Proof. Let f be a local equation of D_t. By (3.2), we have $w_t \ge 2t \log|f| - C$. Since $b_m = 2mt$, every local section $f^{b_m}a$ of $\mathcal{O}(-b_m D_t)$ satisfies

$$|f^{b_m}a|^2 e^{-2mw_t} \le e^{2mC}|a|^2,$$

which is locally integrable. Hence $\mathcal{O}(-b_m D_t)^{\mathrm{an}} \subseteq \mathcal{J}(mw_t)$. Moreover, w_t is continuous on U, so $\mathcal{J}(mw_t)|_{U^{\mathrm{an}}} = \mathcal{O}_{U^{\mathrm{an}}}$.

The multiplier ideal is coherent [3, Proposition 5.7]. By projective GAGA [7, no. 12, Théorèmes 1–3], it algebraizes, together with its inclusions, to a coherent ideal $\mathcal{I}_{t,m}$ satisfying (3.5).

Principalization in characteristic zero [1, Example 1.19(1)] gives a projective modification $\rho_{t,m}$ with smooth source, isomorphic over U, on which the image ideal is invertible. Since $\mathcal{I}_{t,m}|_U = \mathcal{O}_U$, its divisor $F_{t,m}$ is effective and supported outside U. □

Proposition 3.5 (The divisor of a principalized multiplier ideal)**.** *Let X_t, D_t, B_t, w_t, m and $\mathcal{I}_{t,m}$ be as in Proposition 3.4. Let $\rho : Y \to X_t$ be* any *projective principalization with Y smooth that is an isomorphism over U, and let F be the effective Cartier divisor defined by the image ideal*

$$\mathcal{I}_{t,m}\mathcal{O}_Y = \mathcal{O}_Y(-F).$$

Then

$$0 \le \frac{F}{m} \le 2t\rho^*D_t \tag{3.7}$$

as $\mathbb{Q}$-Cartier divisors on Y.

Proof. Fix $y \in Y$ and put $x = \rho(y)$. Choose a local equation f of D_t at x, and local generators $a_1, \dots, a_s$ of $\mathcal{I}_{t,m,x}$. The first inclusion in (3.5) means that there are $c_i \in \mathcal{O}_{X_t,x}$ with

$$f^{b_m} = \sum_{i=1}^{s} c_i a_i, \qquad b_m = 2mt.$$

Apply the local ring homomorphism $\rho^{\#} : \mathcal{O}_{X_t,x} \to \mathcal{O}_{Y,y}$. Its image gives

$$(\rho^{\#} f)^{b_m} \in \mathcal{I}_{t,m}\mathcal{O}_{Y,y}.$$

This uses the image ideal and requires no flatness assumption on ρ.

Choose a nonzero local generator g of $\mathcal{O}_Y(-F)$. The membership above gives

$$(\rho^{\#} f)^{b_m} = g h_y \tag{3.8}$$

for a regular function h_y on a sufficiently small neighborhood. Since ρ is dominant, $h_y \neq 0$, and hence

$$b_m \rho^* D_t - F = \operatorname{div}(h_y) \geq 0.$$

Together with $F \geq 0$, this proves $0 \leq F \leq b_m \rho^* D_t$. Dividing by m gives (3.7). □

3.4 Global generation

Theorem 3.6 (Nadel vanishing and Castelnuovo–Mumford regularity)**.** *Let X be a smooth projective complex variety.*

(i) *If an integral line bundle T carries a singular Hermitian metric h_T with curvature bounded below by a smooth positive form, then*

$$H^i(X, K_X \otimes T \otimes \mathcal{J}(h_T)) = 0 \quad \text{for every } i > 0.$$

The ideal is defined from the local weights of h_T as in Definition 2.8.

(ii) *If H_0 is very ample and a coherent sheaf $\mathcal{S}$ satisfies $H^i(X, \mathcal{S} \otimes H_0^{-i}) = 0$ for every $i > 0$, then $\mathcal{S}$ is globally generated.*

The first assertion is analytic Nadel vanishing [2, Theorem 5.9]; see also [6]. The second is Castelnuovo–Mumford regularity [8, Tag 08A8], applied to the embedding defined by H_0. Projective GAGA identifies analytic and algebraic coherent cohomology.

Lemma 3.7 (Generation after an ample perturbation)**.** *Fix rational $t, \varepsilon > 0$, the line bundle B_t with its semipositive metric from Proposition 3.2, and an ample line bundle A_t as in Lemma 3.3. Give A_t a smooth positive metric and set*

$$E_{t,\varepsilon} = B_t + \varepsilon A_t.$$

For all sufficiently large integers m divisible enough that $mE_{t,\varepsilon}$ and mB_t are integral and $mt \in \mathbb{Z}$, the coherent sheaf

$$\mathcal{S}_{t,\varepsilon,m} = \mathcal{O}_{X_t}(mE_{t,\varepsilon}) \otimes \mathcal{I}_{t,m} \tag{3.9}$$

is globally generated. The threshold depends on the fixed data.

Proof. The product metric on $E_{t,\varepsilon}$ has curvature at least $\varepsilon\omega_{A_t} > 0$. Put $n = \dim X_t$. Choose an integral very ample line bundle H_0 with a smooth positive metric, and a smooth metric on K_{X_t}. For $1 \leq i \leq n$, consider

$$T_{m,i} = mE_{t,\varepsilon} - K_{X_t} - iH_0.$$

On the fixed compact variety X_t, there is a constant C such that

$$c_1(K_{X_t}) + n\omega_{H_0} \le C\omega_{A_t}.$$

The induced singular metric on $T_{m,i}$ therefore satisfies

$$c_1(T_{m,i}) \ge (m\varepsilon - C)\omega_{A_t} > 0$$

for $m\varepsilon > C$. Its weights differ from mw_t by smooth functions, so its multiplier ideal is $\mathcal{I}^{\mathrm{an}}_{t,m}$.

Nadel vanishing and GAGA give

$$H^i(X_t, \mathcal{O}(mE_{t,\varepsilon} - iH_0) \otimes \mathcal{I}_{t,m}) = 0 \qquad (1 \le i \le n).$$

For $i > n$, cohomology vanishes by dimension. Castelnuovo–Mumford regularity now implies that $\mathscr{S}_{t,\varepsilon,m}$ is globally generated. □

3.5 Completion of the proof

Proof of Theorem 1.1 and remark 1.4. Fix X_0, D_0 and an ample $\mathbb{Q}$-line bundle A on X_0. Let h be a compactified metric on $\mathcal{L}$ with $c_1(L_U, h) \ge 0$. Assume first that $D_0 \ne 0$, and fix a rational $\varepsilon > 0$.

Construction of the model line bundle. For a rational $t > 0$, choose a projective modification $p_t : X_t \to X_0$ with X_t smooth and approximation data satisfying (1.1). By Proposition 3.2, the line bundle B_t admits a singular semipositive metric and satisfies

$$0 \le B_t - \mathcal{L} \le 2tD_0.$$

Choose a rational $0 < \lambda \le t/(1+\varepsilon)$. By Lemma 3.3, there exists an ample $\mathbb{Q}$-line bundle A_t on X_t such that

$$-\lambda D_t \le A_t - p_t^* A \le \lambda D_t.$$

Equip A_t with a smooth positive metric.

With these choices fixed, take a sufficiently large divisible integer m satisfying the conditions of Lemma 3.7. By Proposition 3.4, there is a projective modification $\rho = \rho_{t,m} : Y_{t,m} \to X_t$ with $Y_{t,m}$ smooth, such that $\mathcal{I}_{t,m}\mathcal{O}_{Y_{t,m}} = \mathcal{O}_{Y_{t,m}}(-F_{t,m})$. For simplicity, write $F = F_{t,m}$ and $q = p_t \circ \rho$.

Define

$$N_{t,\varepsilon,m} = \rho^*(B_t + \varepsilon A_t) - \frac{F}{m}. \tag{3.10}$$

Global generation and restriction to U. Set

$$V_m = H^0(X_t, \mathscr{S}_{t,\varepsilon,m}).$$

By Lemma 3.7, the evaluation map

$$V_m \otimes \mathcal{O}_{X_t} \twoheadrightarrow \mathscr{S}_{t,\varepsilon,m}$$

is surjective. Pulling back to $Y_{t,m}$ and passing to the image ideal $\mathcal{I}_{t,m}\mathcal{O}_{Y_{t,m}} = \mathcal{O}_{Y_{t,m}}(-F)$ gives

$$V_m \otimes \mathcal{O}_{Y_{t,m}} \twoheadrightarrow \mathcal{O}_{Y_{t,m}}(m\rho^*(B_t + \varepsilon A_t) - F) = \mathcal{O}_{Y_{t,m}}(mN_{t,\varepsilon,m}).$$

Hence $N_{t,\varepsilon,m}$ is semiample.

Since $\mathcal{I}_{t,m}|_U = \mathcal{O}_U$ and the modifications are isomorphisms over U, the specified identifications for B_t and A_t induce

$$N_{t,\varepsilon,m}|_U \simeq L_U + \varepsilon A|_U.$$

All subsequent boundary comparisons use this identification.

Boundary estimate. Put $D_Y = q^*D_0 = \rho^*D_t$. By Proposition 3.5,

$$0 \le \frac{F}{m} \le 2tD_Y.$$

Together with Proposition 3.2, this gives

$$-2tD_0 \le (\rho^*B_t - \mathcal{L}) - \frac{F}{m} \le 2tD_0$$

in the completed boundary order.

Using (3.10), we have

$$N_{t,\varepsilon,m} - (\mathcal{L} + \varepsilon A) = (\rho^*B_t - \mathcal{L}) - \frac{F}{m} + \varepsilon\rho^*(A_t - p_t^*A).$$

The ample adjustment satisfies

$$-\varepsilon\lambda D_Y \le \varepsilon\rho^*(A_t - p_t^*A) \le \varepsilon\lambda D_Y.$$

Consequently,

$$\|N_{t,\varepsilon,m} - (\mathcal{L} + \varepsilon A)\|_{D_0} \le 2t + \varepsilon\lambda \le 3t. \tag{3.11}$$

This estimate holds for every sufficiently large admissible m.

Convergence and nefness. Given $\delta > 0$, choose a rational $0 < t < \delta/3$ and construct $N_{t,\varepsilon,m}$ as above. Then (3.11) proves (1.3).

To obtain a convergent presentation, choose $t_j \downarrow 0$ and an admissible m_j for each t_j. By the triangle inequality,

$$\left\|N_{t_j,\varepsilon,m_j} - N_{t_k,\varepsilon,m_k}\right\|_{D_0} \le 3t_j + 3t_k.$$

Thus the semiample model line bundles N_{t_j,ε,m_j} form a boundary-Cauchy sequence converging to $\mathcal{L} + \varepsilon A$. It follows that $\mathcal{L} + \varepsilon A$ is strongly nef for every rational $\varepsilon > 0$.

For any integer $a \ge 1$, taking $\varepsilon = 1/a$ and multiplying the resulting presentation by a shows that $a\mathcal{L} + A$ is strongly nef. Since A is a fixed strongly nef line bundle, independent of a, Definition 2.4 implies that $\mathcal{L}$ is nef.

If U is projective, then $D_0 = 0$ and $\mathcal{L} = L_U$. For every rational $\varepsilon > 0$, the line bundle $L_U + \varepsilon A$ has a continuous metric with strictly positive curvature. The Nadel–regularity argument of Lemma 3.7, with trivial multiplier ideals, shows that it is semiample. Hence $\mathcal{L}$ is nef in this case as well. □

4 Approximation over a number field

4.1 Arithmetic adelic presentations

Proposition 4.1 (Compatibility of geometric and complex realizations)**.** *Let K be a number field, and let U/K be smooth, geometrically integral, and quasi-projective. For any arithmetic adelic $\mathbb{Q}$-line bundle $\overline{\mathcal{L}} \in \widehat{\mathrm{Pic}}(U/\mathbb{Z})_{\mathbb{Q}}$ and any embedding $\sigma : K \hookrightarrow \mathbb{C}$, its prescribed continuous metric h_σ is a compactified metric on the geometric part $(\mathcal{L}^{\mathrm{geom}})_\sigma$. No positivity or integrability assumption on $\overline{\mathcal{L}}$ is required.*

Proof. This follows from the geometric-part construction [10, Section 2.5.5] and the convergence of model metrics in the proof of [10, Proposition 3.4.1, Section 3.4.2], together with the independence of the boundary topology [10, Section 2.4]. □

4.2 Specialization and descent

Proposition 4.2 (Approximation over the field of definition)**.** *Let $k \subset \mathbb{C}$ be a field of characteristic zero, let X/k be smooth, projective, and geometrically integral, and let $U \subset X$ be a nonempty open subset. Let D be an effective integral Cartier divisor with $\operatorname{Supp} D = X \setminus U$, and let $Q \in \operatorname{Pic}(X)_{\mathbb{Q}}$. The case $U = X$ is allowed, with $D = 0$.*

Suppose that, for some rational $\delta > 0$, there are a projective modification $p_{\mathbb{C}} : Y_{\mathbb{C}} \to X_{\mathbb{C}}$, with $Y_{\mathbb{C}}$ normal and $p_{\mathbb{C}}$ an isomorphism over $U_{\mathbb{C}}$, and a semiample $\mathbb{Q}$-line bundle $N_{\mathbb{C}}$ on $Y_{\mathbb{C}}$, with a specified identification $N_{\mathbb{C}}|_{U_{\mathbb{C}}} \simeq Q|_{U_{\mathbb{C}}}$, such that

$$-\delta p_{\mathbb{C}}^* D \le N_{\mathbb{C}} - p_{\mathbb{C}}^* Q \le \delta p_{\mathbb{C}}^* D. \tag{4.1}$$

Then there exist a projective modification $p : Z \to X$ over k, with Z smooth and geometrically integral and p an isomorphism over U, and a semiample $\mathbb{Q}$-line bundle N_0 on Z, with a specified identification $N_0|_U \simeq Q|_U$, such that

$$-\delta p^* D \le N_0 - p^* Q \le \delta p^* D. \tag{4.2}$$

The inequalities are inequalities of comparison divisors determined by the specified identifications over U.

Proof. Choose a positive integer b clearing all denominators and realizing the open isomorphism, such that $H_{\mathbb{C}} = bN_{\mathbb{C}}$ is globally generated. The inequalities (4.1) give regular maps

$$p_{\mathbb{C}}^* \mathcal{O}_{X_{\mathbb{C}}}(bQ_{\mathbb{C}} - b\delta D_{\mathbb{C}}) \longrightarrow H_{\mathbb{C}} \longrightarrow p_{\mathbb{C}}^* \mathcal{O}_{X_{\mathbb{C}}}(bQ_{\mathbb{C}} + b\delta D_{\mathbb{C}}),$$

whose restrictions to $U_{\mathbb{C}}$ are determined by the specified identification and the canonical boundary trivialization.

By spreading out [8, Tags 01ZM, 01ZR, and 0B8W], the projective modification, line bundle, generating sections, specified isomorphisms over U, and comparison maps are defined over a finitely generated integral k-subalgebra of $\mathbb{C}$. After localization, specialize at a closed point with residue field ℓ, finite over k. Replace the fiber by the normalization of the reduced closure of U_ℓ; this gives a projective modification $Y_\ell \to X_\ell$, unchanged over U_ℓ [8, Tags 0BXR and 0GK4]. The generating sections and comparison maps pull back, and the latter remain nonzero on U_ℓ. Thus the resulting globally generated line bundle H_ℓ satisfies the same bounds, with tolerance $b\delta$.

Choose a finite Galois extension F/k containing ℓ, with group G and degree r. Let W_F be the normalization of the closure of the diagonal U_F in

$$\prod_{\tau \in G,\, X_F} {}^{\tau}Y_F,$$

and let $r_\tau : W_F \to {}^{\tau}Y_F$ be the projections. Permutation of the factors gives a Galois descent datum. The tensor product of the conjugate pullbacks of an auxiliary very ample line bundle on Y_F gives a compatible ample line bundle on W_F. By Galois descent and descent of ampleness [8, Tags 0CCJ, 0CDQ, and 0D2P], W_F descends to a projective modification $p : Z \to X$ with Z normal, isomorphic over U. The same descent applies to

$$\mathscr{B}_F = \bigotimes_{\tau \in G} r_\tau^*({}^{\tau}H_F),$$

giving a line bundle $\mathscr{B}$ on Z with a specified identification $\mathscr{B}|_U \simeq brQ|_U$. It is globally generated by [8, Tag 0B57].

Tensoring and descending the comparison maps gives

$$-br\delta p^*D \le \mathscr{B} - brp^*Q \le br\delta p^*D.$$

Hence

$$N_0 = \frac{1}{br}\mathscr{B} \tag{4.3}$$

is semiample and satisfies (4.2) with the required identification over U. Finally, resolve Z without changing U [1, Theorem 6.1] and pull back N_0. The resolved projective model $U \subset Z$ is geometrically integral because it is birational to X. □

4.3 Proof of the number-field theorem

Proof of Theorem 1.2. Choose a smooth projective K-compactification $U \subset X_0$, an effective integral Cartier divisor D_0 with support $X_0 \setminus U$, and an ample $\mathbb{Q}$-line bundle A on X_0. Assume first that $D_0 \neq 0$. Fix rational $\varepsilon > 0$ and real $\delta > 0$, and choose rational $t > 0$ with $3t < \delta$. Choose a model presentation Q of L_U over K with $\|Q - \mathcal{L}\|_{D_0} < t$. After resolution, we may suppose that Q is defined on a smooth projective model $U \subset X$, with a modification $p : X \to X_0$; put $D_X = p^*D_0$.

By Remark 1.4, there is a semiample complex model line bundle $N_{\mathbb{C}}$, with the prescribed restriction to $L_{U,\sigma} + \varepsilon A_\sigma|_{U_\sigma}$, such that

$$\|N_{\mathbb{C}} - (\mathcal{L}_\sigma + \varepsilon A_\sigma)\|_{D_0} < t.$$

On a common projective model $U_\sigma \subset W$ dominating X_σ, the comparison divisor therefore satisfies

$$-2tD_W \le N_{\mathbb{C},W} - (Q + \varepsilon p^*A)_{\sigma,W} \le 2tD_W.$$

Here D_W is the full pullback of D_0. Applying Proposition 4.2 with reference line bundle $Q + \varepsilon p^*A$ and tolerance $2t$ gives a projective modification $r : Y \to X$ over K with Y smooth and a semiample line bundle N, with the required identification over U, such that

$$-2tr^*D_X \le N - r^*(Q + \varepsilon p^*A) \le 2tr^*D_X.$$

Consequently,

$$\begin{aligned}\|N - (\mathcal{L} + \varepsilon A)\|_{D_0} &\le \|N - r^*(Q + \varepsilon p^*A)\|_{D_0} + \|Q - \mathcal{L}\|_{D_0} \\ &< 3t < \delta.\end{aligned} \tag{4.4}$$

Letting δ tend to zero gives a boundary-Cauchy sequence of semiample model line bundles converging to $\mathcal{L} + \varepsilon A$, which is therefore strongly nef. For integers $a, j \ge 1$, take $\varepsilon = 1/a$ and $\delta = 1/(aj)$, and multiply the approximating line bundle by a. The resulting semiample line bundle $N_{a,j}$ has the specified restriction $aL_U + A|_U$ and

$$\|N_{a,j} - (a\mathcal{L} + A)\|_{D_0} < j^{-1}. \tag{4.5}$$

Thus $a\mathcal{L} + A$ is strongly nef for every $a \ge 1$. Since A is fixed and strongly nef, $\mathcal{L}$ is nef.

If U is projective, then $\mathcal{L} = L_U$. The projective case of Remark 1.4 and descent of global generation [8, Tag 0B57] imply that $L_U + \varepsilon A$ is semiample over K, so the same conclusion holds with zero boundary error. The assertion for arithmetic adelic line bundles follows from Proposition 4.1. □

Remark 4.3 (Semiample approximation over K). Under the hypotheses of Theorem 1.2, fix a smooth projective K-compactification $U \subset X_0$, an effective integral Cartier divisor D_0 with support $X_0 \setminus U$, and an ample $\mathbb{Q}$-line bundle A on X_0. For every rational $\varepsilon > 0$ and every real $\delta > 0$, the proof gives a projective modification $q : Y \to X_0$ over K, with Y smooth and q an isomorphism over U, and a semiample $\mathbb{Q}$-line bundle N on Y, with a specified isomorphism

$$L_U + \varepsilon A|_U \xrightarrow{\sim} N|_U,$$

such that

$$\|N - (\mathcal{L} + \varepsilon A)\|_{D_0} < \delta.$$

Thus $\mathcal{L} + \varepsilon A$ is strongly nef for every rational $\varepsilon > 0$. Taking $\varepsilon = 1/a$ and scaling by a shows that $a\mathcal{L} + A$ is strongly nef for every integer $a \geq 1$, with the same A; the approximations satisfy (4.5). If U is projective, one may take $Y = X_0 = U$ and $N = L_U + \varepsilon A$.

5 Proof of the Gross–Schoen theorem

Let $M, \mathcal{C}, J$, and $\overline{G}_g$ be as in Theorem 1.3 and definition 2.12.

Proposition 5.1 (The height and curvature of $\overline{G}_g$)**.** *The adelic line bundle $\overline{G}_g$ is integrable. For every algebraic point s of M, its normalized point height is*

$$h_{\overline{G}_g}(s) = \langle \Delta_{\xi_s}, \Delta_{\xi_s} \rangle_{\mathrm{BB}}.$$

At every complex embedding $\sigma : K \hookrightarrow \mathbb{C}$, its prescribed metric is smooth and satisfies

$$c_1(\overline{G}_{g,\sigma}) = \beta_{\mathrm{GS}} \geq 0, \tag{5.1}$$

where β_{GS} is the Betti form of the Gross–Schoen normal function, with the Poincaré normalization in Definition 2.12.

Proof. The construction in [5, Section 4.1] gives an integrable adelic Poincaré line bundle. Pullback and the relative Deligne pairing preserve integrability by [10, Section 2.5.5 and Theorem 4.1.3]. The height identity is [5, Theorem 4.1]. The smoothness and curvature identity are [5, Section 4.2 and Theorem 4.2], where the metric is related locally to the metrized biextension associated with the Gross–Schoen normal function. These identities pull back to M and apply to (1.2). □

Proof of Theorem 1.3. By Proposition 5.1, the prescribed metric of $\overline{G}_g$ at a complex embedding is smooth and has semipositive first Chern current. The arithmetic adelic case of Theorem 1.2 gives the nefness of G_g^{geom}. □

Remark 5.2 (Approximation of the Gross–Schoen line bundle). Let $G_{g,M}$ denote the underlying line bundle of $\overline{G}_g$ on M. Fix a smooth projective K-compactification $M \subset X_0$, an effective integral Cartier divisor D_0 with support $X_0 \setminus M$, and an ample $\mathbb{Q}$-line bundle A on X_0. For every pair of integers $a, j \geq 1$, Remark 4.3 gives a projective modification $q_{a,j} : Y_{a,j} \to X_0$ over K, with $Y_{a,j}$ smooth and $q_{a,j}$ an isomorphism over M, and a semiample $\mathbb{Q}$-line bundle $N_{a,j}$, with a specified isomorphism

$$aG_{g,M} + A|_M \xrightarrow{\sim} N_{a,j}|_M,$$

such that

$$\left\| N_{a,j} - (aG_g^{\mathrm{geom}} + A) \right\|_{D_0} \leq j^{-1}. \tag{5.2}$$

Here one takes $\varepsilon = 1/a$ and $\delta = 1/(aj)$ and multiplies the approximation by a. The line bundle A is fixed for all a and j.

Nefness is preserved by algebraic pullback to smooth quasi-projective varieties [10, Section 2.5.5]. It is also preserved by extension of the ground field, by base change of the approximations in (5.2).

Index of notation

Symbol	Meaning	Reference
U, X_0, D_0	Open variety, fixed compactification, and full Cartier boundary.	Section 2.1
$D_X = p^* D_0$	Boundary divisor on a model $U \subset X$, pulled back along p.	(2.1)
L_U, α_P	Underlying $\mathbb{Q}$-line bundle and its specified isomorphism with a model restriction.	Definition 2.1
$\mathcal{L}, P$	Compactified line bundle and a model line bundle, also viewed as its constant class.	Definition 2.3
$\widehat{\mathrm{Pic}}(U/k)_{\mathbb{Q}}$	Group of compactified $\mathbb{Q}$-line bundles over a field k.	Definition 2.3
$\Delta(Q, P)$	Boundary divisor determined by the specified comparison over U.	(2.2)
$\|\cdot\|_{D_0}$	Boundary norm on model differences and their completion.	Definition 2.2; (2.4)
$h, \varphi_{h,e}$	Metric and local weight $-\log \|e\|_h$.	Definition 2.5
$\mathcal{L}, h$	Compactified line bundle with a compactified metric.	Definition 2.7
g_0, g_X	Boundary Green function and its pullback.	Definition 2.6
t, P_t, h_t	Approximation tolerance, model line bundle, and model metric.	(1.1)
B_t, w_t	Boundary-corrected line bundle and its extended psh weight.	Proposition 3.2
A, A_t, λ	Fixed ample line bundle, its ample perturbation on a model, and the boundary tolerance.	Lemma 3.3
$\varepsilon, E_{t,\varepsilon}$	Positive rational parameter and $B_t + \varepsilon A_t$.	Lemma 3.7
$\mathcal{J}(mw_t)$	Coherent analytic multiplier ideal sheaf.	Definition 2.8
$\mathcal{I}_{t,m}$	Algebraic ideal corresponding to $\mathcal{J}(mw_t)$.	Proposition 3.4
$\rho_{t,m}, F_{t,m}$	Principalization and effective divisor of the image ideal.	(3.6)
$\mathscr{S}_{t,\varepsilon,m}$	Globally generated sheaf $\mathcal{O}(mE_{t,\varepsilon}) \otimes \mathcal{I}_{t,m}$.	(3.9)
$N_{t,\varepsilon,m}$	Semiample model approximation on $Y_{t,m}$.	(3.10)
$\overline{\mathcal{L}}, \mathcal{L}^{\mathrm{geom}}$	Arithmetic adelic line bundle and its geometric part.	Definitions 2.9 and 2.10
σ, h_σ	Complex embedding and the prescribed metric at that embedding.	Definition 2.10
$\overline{G}^Y_{\mathcal{C}/S}$	Yuan's Gross–Schoen line bundle.	Definition 2.11
$\overline{\mathcal{O}}(\Phi_S)$	Adelic line bundle associated with the relative φ-invariant.	(2.9)
$\overline{G}_g, G_g^{\mathrm{geom}}$	Gao–Zhang's adelic Deligne pairing and its geometric part.	Definition 2.12

Symbol	Meaning	Reference
$G_{g,M}, \beta_{\mathrm{GS}}$	Underlying Gross–Schoen line bundle and its complex curvature form.	Definition 2.12 and proposition 5.1
$Q, N_{\mathbb{C}}, N_0$	Reference line bundle, complex approximation, and approximation over the field of definition.	Proposition 4.2
$\mathscr{B}_F, \mathscr{B}$	Tensor product of the Galois conjugates and its descent; $N_0 = \mathscr{B}/(br)$.	(4.3)

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